



Microwave aided scalable synthesis of sulfur, nitrogen co-doped few-layered graphene material for high-performance supercapacitors

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ABSTRACT

Doping with heteroatoms has become an approach for improving the electrochemical performance of few-layered graphene. In this work, Sulfur-nitrogen co-doped few-layered graphene synthesized from the graphite flakes acid treated with H_2SO_4 and HNO_3 followed by microwave irradiation. Sulfur-nitrogen co-doped few-layered graphene consists of less than 15 graphene layers with a high degree of graphitization. The supercapacitor exhibited a specific energy density of 15 Wh kg^{-1} at a power density of 300 W kg^{-1} at room temperature in aqueous electrolyte. The S, N-FLG electrodes show the specific capacitance of 298 F g^{-1} at a current density of 1 A g^{-1} and stable over 10,000 continuous charge-discharge cycles with 95% capacitance retention at 1 A g^{-1} . The obtained capacitance is due to maximum utilization of few-layered graphene sheets, highest intrinsic surface capacitance due to the synergistic effect of the formation of N-S-H hydrogen bonds and S, N co-doping in graphene aromatic rings.

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1. Introduction

Graphene has attracted attention in the field of energy storage and conversion due its excellent surface area ($2675 \text{ m}^2 \text{ g}^{-1}$), electrical conductivity, chemical stability, mechanical ($\sim 1 \text{ TPa}$), and thermal stability [1–3]. Graphene-based supercapacitors have the specific capacitance in the range from $100\text{--}200 \text{ F g}^{-1}$, which is less than its theoretical specific capacitance of 550 F g^{-1} when the entire surface area is fully utilized [4]. The active sites, functional groups, and surface area of graphene strongly impact the capacitive properties [5,6]. Single or few-layer graphene offer short ion diffusion path and low-contact resistance which is essential for high power delivery of supercapacitors. Recent investigations show the heteroatom (B, N, P, and S)-doped graphenes have a difference in electron negativities of carbon and heteroatoms which provides more electroactive sites than pristine graphene. Heteroatom doped graphenes have gained more interest in recent times due to significant improvement in capacitive properties [7–11].

Compare to a single heteroatom (B, N, P, and S) doping, co-doped (B and N, N, and S and P and N) graphenes exhibits ex-

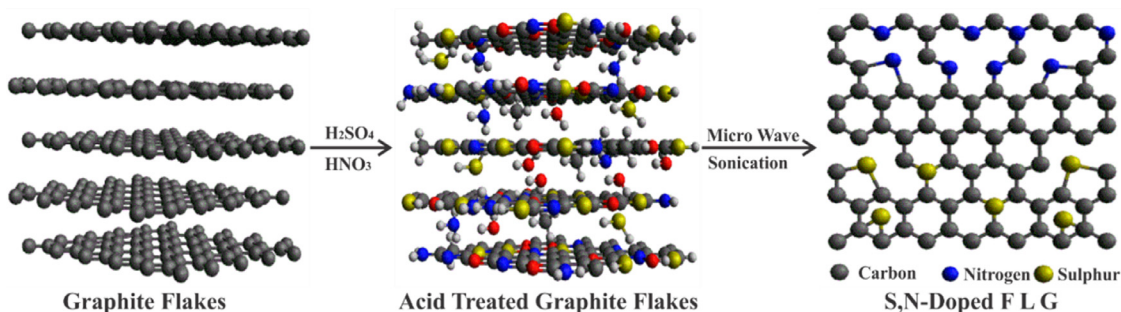
cellent electrochemical properties. It tunes the electrical conductivity, surface active sites, electrode wettability which in turn improves the capacitive properties and bonding in the graphene layers [12,13]. Among all hetero atom-doped graphene's, the S and N atoms doped graphenes have been playing an essential role in the energy storage, due to dopant in graphene. The presence of co-doped heteroatoms in the graphene lattice enhances the activation region is due to the more charge distribution of the carbon atoms and thereby increases the charge storage property, decreases charge transfer resistance and improves the wettability nature [14]. Co-doping provides the electron transfer, it consists of different bonding and different structure in comparison to the single atom doping. Besides, due to the large size of sulfur it provides the easily polarizable electron pairs subsequently shows the more charge storage and high chemical reactivity, nitrogen doping helps to tune the electron transport properties of the carbon across the graphene lattice [15,16].

Several methods for synthesis of S, N co-doped few-layered Graphene (S, N-FLG) have been developed [17,18]. Literature studies reveal that Sulfur functionalized graphene aerogel shows the specific capacitance 1089 F g^{-1} at a current density of 1 A g^{-1} and 860 F g^{-1} at 20 A g^{-1} for 10,000 cycles in 1 M KOH solution. The presence of sulfone groups in the graphene aerogel enhances the electrochemical performances [19]. S incorporated RGO

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Scheme 1. Graphic representation of S, N co-doped few-layered graphene.

aerogel has the 3D structure, it delivers the specific capacitance of 445.6 F g^{-1} at 5 mV s^{-1} . The obtained specific capacitance is due to the π - π stacking between the RGO layers and it forms the hydrogen bonding between S-H groups [20]. Nitrogen modified few-layer graphene exhibits the capacitance of 227 F g^{-1} at a current density of 1 A g^{-1} is due to the quaternary nitrogen enhances the electron transfer in doped graphene [21]. Low temperature synthesized nitrogen-doped porous graphene exhibits the capacitance of $\sim 145 \text{ F g}^{-1}$ with capacitance retention of 96.4 % at the end of 5,000 charge-discharge cycles at a current density of 10 A g^{-1} . The enhancement in the electrochemical performance is due to the pyridinic nitrogen present in the nitrogen-doped graphene [22]. N and S co-doped graphene in 6 M KOH solution exhibit specific capacitance 566 F g^{-1} at 0.5 A g^{-1} with 95 % capacitance retention (274 F g^{-1} at 3 A g^{-1}) for 2000 cycles. The improvement in the capacitance is due to the synergetic effect of nitrogen and sulfur present in the graphene layers [4].

Existing methods are having drawbacks such as long reaction times, exclusive instruments, and restricted to scalability. To overcome these limits, a feasible method is anticipated to prepare S, N-FLG. In this perspective, the method contains intercalation and oxidation of graphite with acids followed by microwave irradiation have been made noteworthy development in the preparation of S, N-doped FLG in a viable manner [23,24].

Herein, we reveal a synthesis of S, N-FLG from graphite flakes acid medium subsequently microwave irradiation for exfoliation in for a few seconds and study their electrochemical performance in supercapacitors. Our synthesis method is a commercially viable, fast approach, and scalable.

2. Experimental

2.1. Synthesis of S, N co-doped few-layered Graphene (S, N-FLG)

Sulfur-nitrogen co-doped few-layered graphene (S, N-FLG) was synthesized from the H_2SO_4 and HNO_3 acid-treated graphite flakes (ATGF) by microwave irradiation as demonstrated in Scheme 1. In brief, the graphite flakes ($\sim 150 \mu\text{m}$) stirred overnight in H_2SO_4 and HNO_3 (v/v 3:1) for 15 hr. to achieve uniform intercalation of solvents in graphene layers of graphite. The acid-treated graphite flakes (ATGF) was washed with distilled water several times to remove the supernatant until a neutral pH (7) reached. Finally, the filtrate was dried in a hot air oven at 100°C for overnight. As prepared acid-treated graphite flakes were exposed to microwave irradiation (900W and 2.45GHz) for a few seconds. Graphene layers were obtained upon irradiation due to the exfoliation of individual sheets of ATGF along the basal plane direction with crackling sound and sparkles. These graphene layers were sonicated by ultra-probe in ethanol until to get uniform suspensions of few-layered graphene sheets.

2.2. Structural and physical characterizations

Morphologies of S,N-FLGs were characterized by Field emission scanning electron microscope (FESEM) (Model Zeiss Ultra 55) and transmission electron micrographs (Model FEI Technai G2 S-Twin). Graphite, S,N-FLG were characterized by XRD, and patterns were recorded from 10 to 100° with 0.02° degree/sec using $\text{Cu K}\alpha$ X-ray source ($\lambda=1.54 \text{ \AA}$) (Bruker's AXS Model D8 Advanced System). Micro-Raman spectra of synthesized materials recorded using an Nd-YAG 532 nm laser model alpha 300 of WiTec, Germany. Fourier Transform Infrared (FTIR) spectroscopy was used to understand the composition of the samples. In this study, the FTIR spectra of the samples were collected using a Perkin Elmer FT-IR X 1760 instrument. Spectra were taken in the transmission mode. Changes in the absorption bands were investigated in the 400 - 4000 cm^{-1} region. Brunauer-Emmett-Teller (BET) measurements were carried out on Micromeritics, ASAP2020 surface area analyzer.

X-Ray Photoelectron Spectroscopy (XPS) measurements were performed using VG Scientific ESCALab Mark II X-ray photoelectron spectrometer using a monochromatic $\text{Al K}\alpha$ X-ray source ($\sim 1486.6 \text{ eV}$). Survey spectra were collected over the binding energy range between 0 and 1200 eV using a step size of 1 eV and pass energy of 50 eV . In contrast, the high-resolution spectra of C1s , Al 2p , and Al KLL were collected using a step size of 0.2 eV and pass energy of 20 eV . All binding energies were corrected concerning the C1s peak at 284.5 eV . The spectra were deconvoluted using XPS peak fit 4.0 software based on Gaussian-Lorentzian functions.

2.3. Electrode preparation and electrochemical characterizations

The electrodes were prepared using a graphite sheet as a current collector. Capacitance contribution from the graphite sheet is very negligible. The electrodes were prepared from the composite slurry of 80 % of S, N-FLG, 10% of carbon black (Super C65), 10 % of PVDF binder in N-Methyl-2-pyrrolidone (NMP) using doctor blade technique. The electrodes were dried at 100°C for overnight under vacuum oven. The electrode area was 4 cm^2 ($2 \text{ cm} \times 2 \text{ cm}$) with active material loading is about 0.8 g cm^{-2} . The electrochemical performance of the composite electrode was investigated by using cyclic voltammogram (CV) and galvanostatic charge-discharge tests (GCD). Capacitive behavior was studied by using a three-electrode configuration using S, N-FLG composite electrode as a working electrode, Titanium foil as a counter electrode, and a saturated calomel electrode (SCE) as a reference electrode. CV and charge-discharge tests were carried out the potential range in between -0.6 to 0 V in 6 M NaOH solution. Solartron Analytical 1255 FRA (Frequency response analyzer) (from Solartron analytical, Oak Ridge, TN, USA) instrument was used to conduct all the electrochemical characterizations.

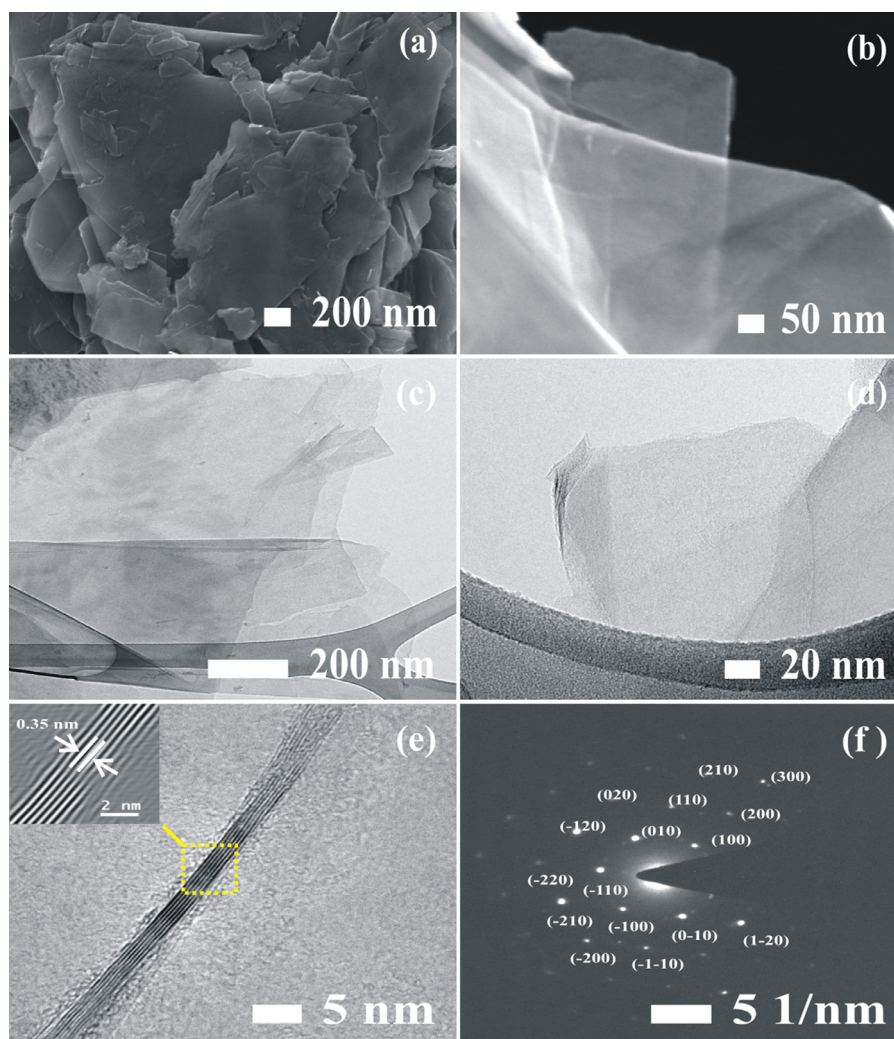


Fig. 1. SEM images of (a) graphite flakes (b) S, N-FLG. TEM images of (c-e) S, N-FLG with different magnifications, (f) SAED pattern S, N-FLG.

3. Results and discussions

3.1. Physical and structural characterizations

The morphology of graphite flakes (GFs), and S, N-FLG observed by FESEM are illustrated in Fig. 1 a and b, respectively. Fig. 1(a) shows graphite flakes are very thick, which consists of a huge number of stacked graphene layers. S, N-FLG have few layers of graphene layers confirmed by transmission electron micrographs shown in Fig. 1 (c-f). The high-resolution micrographs are shown in Fig. 1(e) reveal the number of layers (clearly annotated by arrows) and the thickness of the sheet is about 7 nm and it may vary for different locations. The d spacing as measured from Fig. 1 (e) is ~ 0.35 nm. The SAED pattern shown in Fig. 1 (f) depicts a set of six-fold symmetric diffraction spots, indicating the presence of few-layered graphene and crystalline nature of graphene [26].

X-ray patterns of GFs and S, N-FLG are shown in Fig. 2a. The (002) signature peak of carbon is observed in both cases. The (002) reflection in S, N-FLG, is broad and appears to be at a higher angle (31.6°) compared to graphite (26.2°). The (002) signature peak of carbon in S, N-FLG appears at a higher angle and broad because of difference in aromatics rings of graphene and dopant atoms S and N. Size and change in lattice parameters leads to the peak shift in any kind of materials. In the current case, S and N were incorporated in Graphene in the form of their corresponding groups like

Sulphur appears sulfate/sulfonate groups, Nitrogen appears quaternary nitrogen and bipolaron = NH^+ bond. The diameter of S and N is larger than carbon, and their corresponding groups are also much larger. Consequently, the interlayer distance increases, which leads to the change in peak position and size in the XRD pattern. Because of the minute defects or possibly surface oxidations, the intensity of the D band in Fig. 2b is low in GF compared to S, N-FLG. The prominent D-band in S, N-FLG is due to doping of S and N in FLG. The simultaneous observation of high-intensity G band, low-intensity D band and the single 2D band with an intensity lower than that of G band are an indication of the presence of few-layered graphene [25]. The intensity ratio I_D/I_G which is a measure of disorder in the material was found to be 0.7 indicating a high graphitization degree while I_G/I_{2D} was ~ 1.3 corresponding to the presence of less than 15 graphene layers in S, N-FLG.

The FT-IR spectra in Fig. 2c show no significant peaks for graphite. The presence of different oxygen functionalities in S, N-FLG spectra are detected. The appearance of a peak at 3435 cm^{-1} corresponds to O-H stretching vibrations of COOH, 2362 cm^{-1} peak corresponds to C(O)O stretching vibration, while C=O stretching vibration of COOH groups located at 1714 cm^{-1} . Besides, peaks between $1000\text{--}1300\text{ cm}^{-1}$ are due to oxidized carbons (C-O) of alcohols, acids, esters. CH_2 twist peak observed at 1227 cm^{-1} , and C-N/C-O stretching vibration observed at 1102 cm^{-1} [26,27]. The BET measured for S, N-FLG are $\sim 10.76 (\pm 0.04)\text{ m}^2/\text{g}$. The average pore

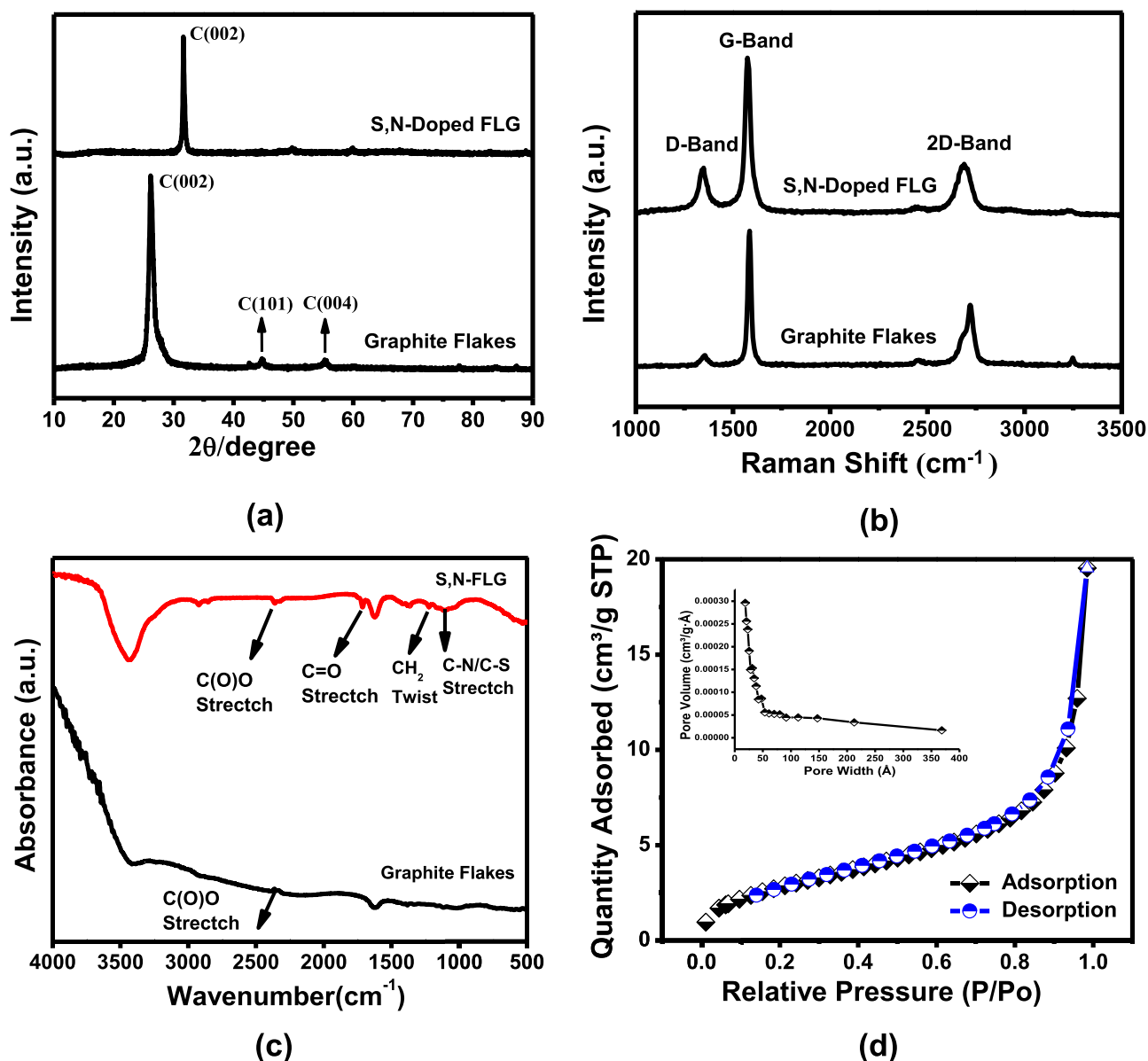


Fig. 2. (a) XRD pattern, (b) Raman spectra, and (c) FT-IR spectra of Graphite flakes and S, N-FLG as indicated in the figures. (d) BET adsorption isotherm of S, N-FLG, the inset is the average pore volume and size plot.

volume and size obtained from BET analysis are $\sim 0.031 \text{ cm}^3/\text{g}$ and $\sim 10.16 \text{ nm}$, respectively (Inset to Fig. 2d). Isotherm of nitrogen gas adsorption and desorption followed the type III isotherm among the BET classification type's I–VI (Fig. 2d).

X-ray photoelectron spectroscopy (XPS) has been used to examine the chemical compositions and element doping in S, N - FLG, as shown in Fig. 3 a-d. To understand the doping and their structure, deconvolution of the original peaks has been considered. S, N-FLG survey spectrum, and atomic percentages as shown in Supplementary Fig. S1 and Table 1. S, N-FLG consists of 72.75% of carbon, 17.57% of oxygen, 5.22% of nitrogen, and 3.84% of sulfur. Fig. 3(a) shows the S2p XPS spectra of S, N-FLG. The peaks at 168.3, 169.3 arising from $(-\text{C}-\text{SO}_x, x=2.3)$ connected to sulfate (from the acid mixture during synthesis) or sulfonate [28–29] groups. The sulfone bridges $(-\text{C}-\text{SO}_2-\text{C}-)$ appeared at 170.1 eV. These sulfone bridges form as a consequence of the oxidizing conditions present during the chemical activation process. The peak at 170.6 eV, is probably due to the chemisorption of oxygen, with the formation of SO_4^{2-} like species [30,31]. S, N -FLG (Fig. 3b) The low B.E peak at 397.2

resembles with N- doped graphene having a majority of nitrogen atoms. The other peaks at 401.1 eV, 402.5 eV correspond to the quaternary nitrogen and bipolaron $=\text{NH}^+$ -bond [32–34].

The O1s spectra as shown in Fig. 3c shows peaks at 531.3, 531.9, 532.4, 532.9, and 533.6 are assigned to $\text{C}=\text{O}$; $\text{C}-\text{OH}$; $\text{C}-\text{O}$ and $\text{C}=\text{O}$; $\text{C}-\text{O}$ doped hetero atom; $\text{O}-\text{C}-\text{O}$ and H_2O , respectively [35–38]. All these peak evidence the presence of functional groups on the surface of the graphene layers.

S, N-FLG the carbon peak in Fig. 3d can be deconvoluted into peaks at 283.8, 284.2, 284.5, 284.8, 286.1, 288.1, 291.4 that are assigned to the C-C, $\text{C}-\text{C}/\text{C}=\text{C}$ in the aromatic rings, Sp^2-C , N-C and N-S, $\text{C}-\text{OH}$ of alkoxy and C-S, $\text{C}=\text{O}$ of carbonyl and carboxylic acid and $\pi-\pi^*$ transition, respectively, as indicated in the Fig. 3d. As the number of functional groups increased in S, N-FLG carbon peak, it becomes asymmetric and narrowed towards low and high binding energy sides. In addition to the Gaussian instrumental broadening, significant asymmetry behaviour of C1s towards higher binding energies is due to excitation of many low-energy electron-hole pairs, which screen the core hole and visible

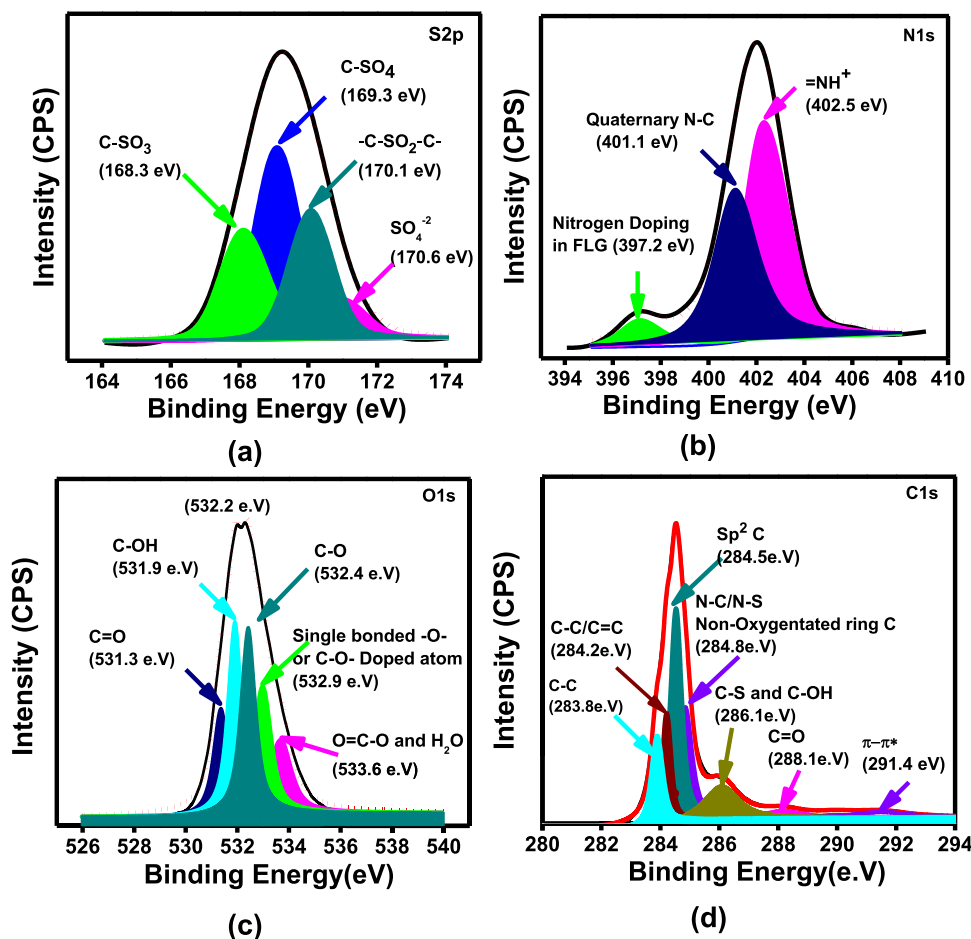


Fig. 3. XPS Spectra of S, N-FLG (a) S2p, (b) N1s, (c) O1s (d) C1s deconvoluted spectra.

as higher binding energies [39]. This asymmetric line shape can be described in the Doniach–Šunjić form as mentioned in the literature [39]. In XPS studies, each deconvoluted peak of carbon represents the respective bonding with sulfur groups and others. These S incorporation into sp² network of graphene mainly contribute to the asymmetric and narrowed towards low and high binding energy sides in XPS [40,41].

S, N-FLG suggesting the restoration of the π -electronic conjugated network structure after reduction. The band at 284.2 indicating carbon-rich material and lower oxygen peaks recognized due to impurities removal from ambient air oxidation or acid treatment during synthesis. These observations are well coinciding with the FTIR results of S, N-FLG [42–45].

3.2. Electrochemical performance of S, N-FLG as electrode material for supercapacitor

CV plots of S, N-FLG composite electrode at low scan rates of 5 and 10 mV s⁻¹ show nearly rectangular shape (Fig. 4 a, b). As the scan rate increases from 10 to 100 mV s⁻¹ rectangular shape disappears and converts into the quasi-rectangular shape due to kinetics limitations at high scan rate (Fig. 4b). Besides, at high scan rate, a smaller number of electrolyte ions are assessed into the bulk of the electrode. CV curves at all scan rates exhibit some deviation (shape of the curve) from the capacitive behavior due to the pseudocapacitive nature of S, N hetero atoms doped into the few-layer graphene. Graphene layer edges consist of the pyridinic nitrogen and contributes an appreciable amount of pseudocapacitance to

the total capacitance. The presence of sulfone and sulfoxide groups in FLG also improves the total capacitance.

Charge storage behavior of S, N-FLG electrodes studied by using Power-law from the CVs at different scan rates as shown in Eq. (1).

$$i = av^b \quad (1)$$

Where a and b are the adjustable parameters, when $b = 0.5$ reveals the solid phase diffusion-controlled process, $b = 1$ reveals the charge storage is a surface-confined charge transfer process [46]. b value obtained from the linear fit between the $\log v$ vs $\log i$ as shown in Fig. 4c. b value of close to 0.5 (Supplementary Table 2) obtained from the plot of $\log v$ vs $\log i$ at different voltages indicates the charge storage process consists of both diffusion-controlled processes as well as surface-confined charge transfer process [47,48].

The diffusion coefficient was calculated by the Randles Sevcik equation.

$$i_p = 2.69 \times 10^5 n^{3/2} A D_0^{1/2} v^{1/2} C_0 \quad (2)$$

Here i_p shows the peak current density, n is the number of electrons transferred, A reveals the area of the electrode, D_0 is diffusion coefficient, v is the scan rate and C_0 is the concentration of electrolyte. The linear plot between peak current vs square root of the scan rate ($v^{1/2}$) as shown in Fig. 4d. The anodic, cathodic peak current density shows the diffusion of hydroxyl ions, sodium ions respectively. The diffusion coefficients of D_{Na^+} , D_{OH^-} are the 3.31×10^{-11} , 3.29×10^{-11} cm² s⁻¹, respectively in 6M NaOH solution. The obtained results show the diffusion coefficient values are

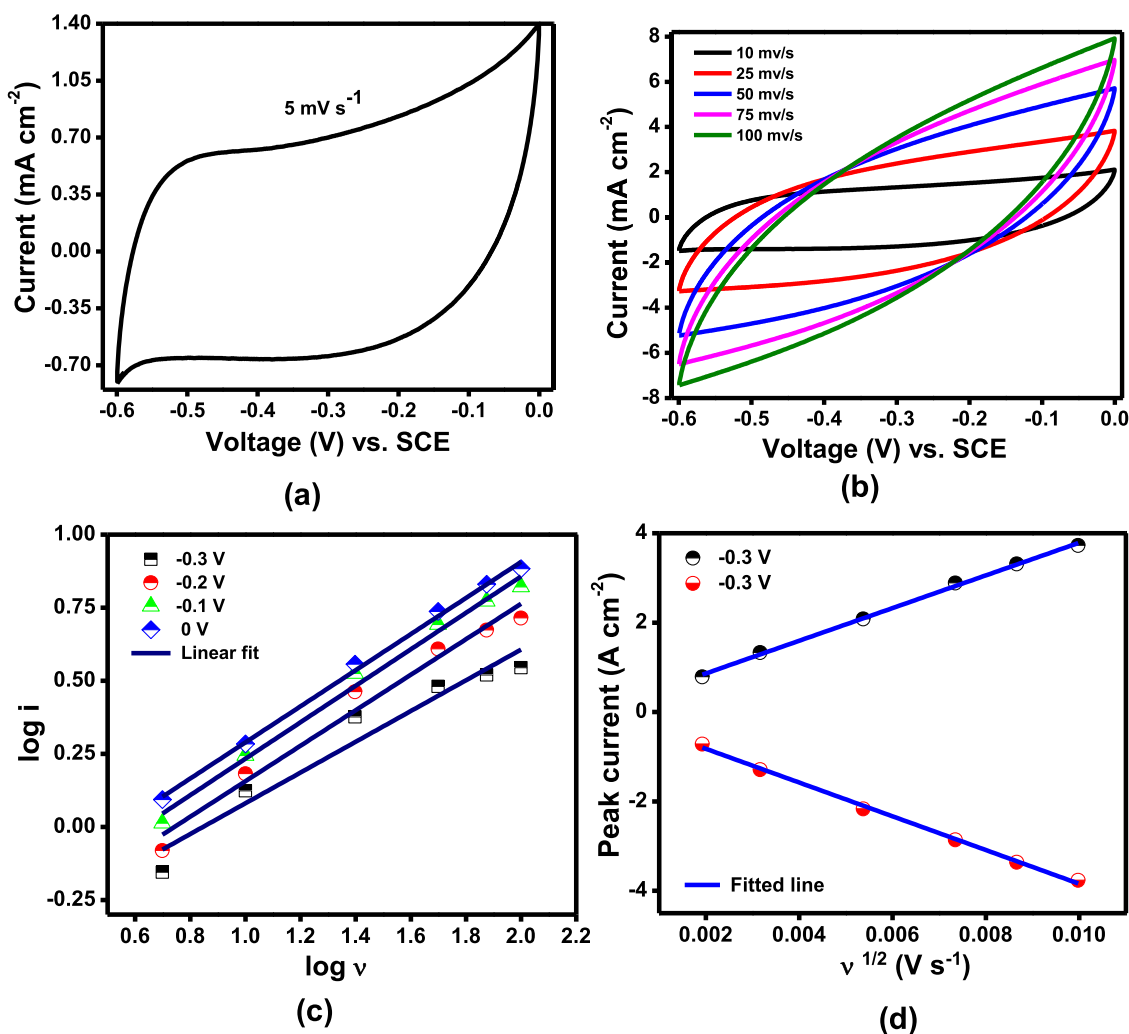


Fig. 4. Cyclic voltammograms curve of S, N-FLG (a) at scan rate of 5 mV s⁻¹, (b) various scan rates (10, 25, 50, 75 and 100 mV S⁻¹), (c) Linear pot of log v vs Log i, (d) Linear fit between the peak current density vs square root of scan rate.

nearly equal, it reveals the cations, as well as anions, are equally penetrating to the electrode.

The specific capacitances of 298, 288, 250, 226 and 200 F g⁻¹ at current densities of 1, 2, 3, 4 and 5 A g⁻¹, respectively are obtained for S, N-FLG electrodes in 6M NaOH solution in the potential range in between -0.6 to 0 V as shown in Fig. 5 (a,b). Specific capacitance calculated from the following formula; $C_{sp} = \frac{(I\Delta t)}{(m\Delta V)}$. Galvanostatic charge-discharge curves exhibit non-linear and nearly triangular shape at all current densities. Non-linear shape arises from the pseudocapacitance of sulfone and sulfoxide groups in S, N-FLG which is faradic. As the current density increases from 1 to 5 A g⁻¹ the specific capacitance value decreases is due to IR drop, a smaller number of ions diffused into the interior of the electrode, and less utilization of the active material. IR (Ohmic) drop is due to the pseudocapacitive behavior containing the sulfone and sulfoxide groups in graphene layers [14]. Few layered graphenes with less agglomeration improves the wettability in the aqueous electrolyte and boosts the capacitance at high current densities of 3 and 5 A g⁻¹. Besides electronic properties are enhanced due to presence of N and S and apart from the graphene layers.

Cycling stability plays an imperative role in supercapacitors for practical applications. S, N-FLG composite electrode shows good cycling stability of 10000 cycles at 1 A g⁻¹ as shown in Fig. 5c. S, N-FLG electrode sustains the almost 95% of capacitance retention during 10000 continuous charge-discharge cycles.

S, N-FLG electrode Ragone plots derived from the galvanostatic charge-discharge curves as shown in Fig. 5d. S, N-FLG composite electrode energy density and power density was calculated from galvanostatic charge-discharge curves at different current densities as per the following equations [3,4].

$$E = \frac{C(\Delta V)^2}{2} \quad (3)$$

$$P = \frac{E}{t} \quad (4)$$

Where E, P are the energy density (Wh kg⁻¹) and power density (W kg⁻¹), C is the specific capacitance, ΔV is the voltage difference, t is the time in seconds. S, N-FLG electrode shows an energy density of 15 Wh kg⁻¹ at a power density of 300 W kg⁻¹ and about 10 Wh kg⁻¹ at 1715 W kg⁻¹. Comparison of energy density and power density of different materials obtained through different synthesis routes as shown in supplementary Table 3.

4. Discussions

S, N-FLG composite electrodes electrochemical performance is due to the synergetic effect of N and S incorporated in the graphene lattice which boosts the electrochemical utilization. The advantage of the electrode is due to the presence of nitrogen in the

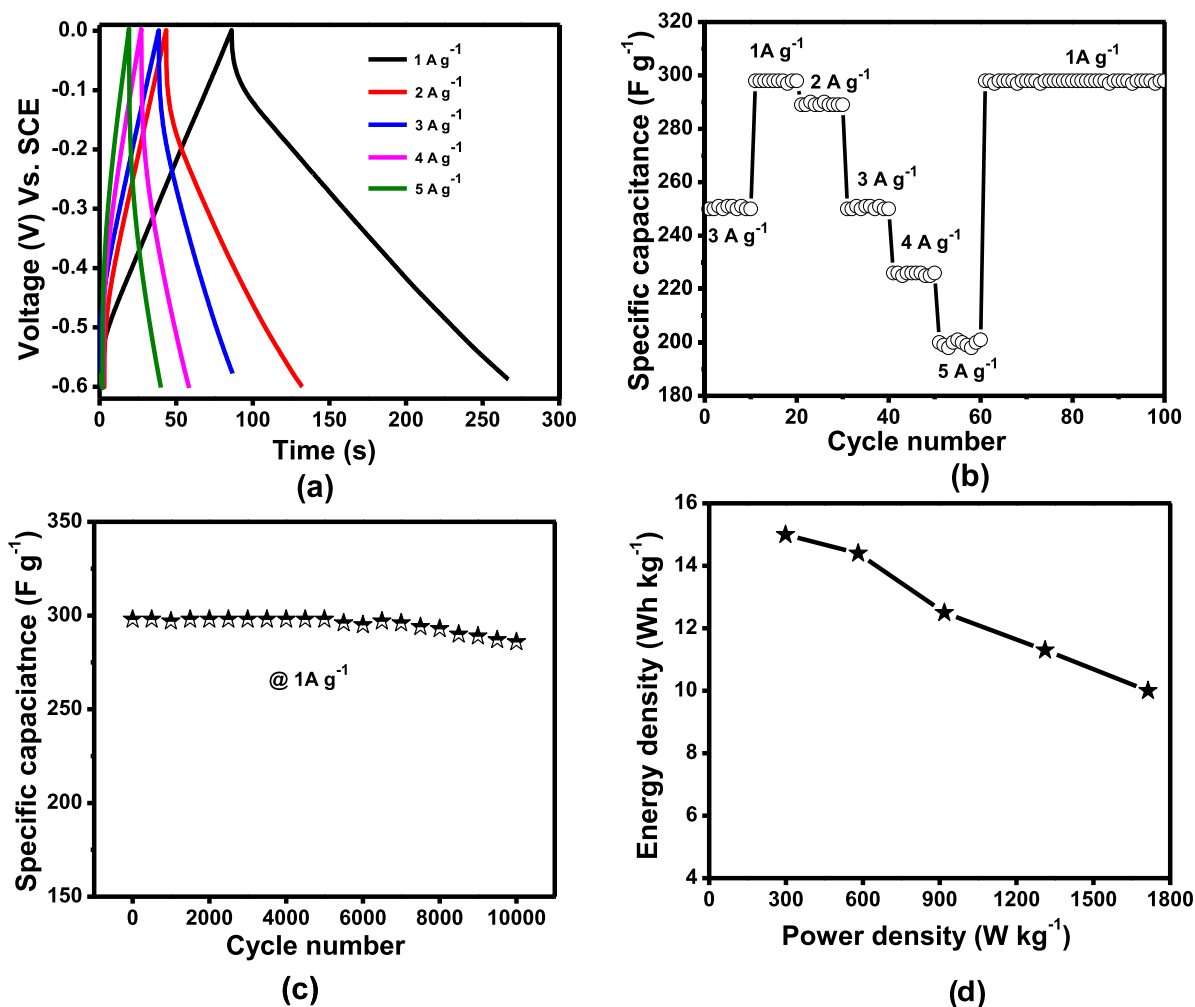
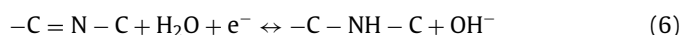
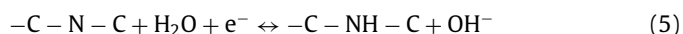


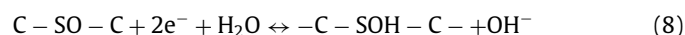
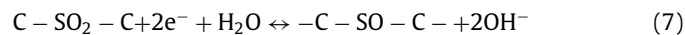
Fig. 5. (a) Galvanostatic charge-discharge curves at different current densities (1,2,3,4 and 5 A g⁻¹) of S, N-FLG, (b) Rate performance of S, N-FLG at different current densities shown in the Fig., (c) Cycling performance of S, N-FLG at 1 A g⁻¹ in the voltage range between the -0.6 to 0 V, (d) Ragone plot (Energy density vs. Power density) of S, N-FLG

graphene layers forms the pyrrolic, graphitic, and pyridine bonds. The pyridinic nitrogen formed at the edge of the graphene layers and more contribution to capacitive behavior. Few layer graphene reduces agglomeration and Pyrrolic nitrogen contained FLG exhibits the wettability graphene in the aqueous electrolyte. The presence of electronegative nature of nitrogen in the FLG exhibits the different configurations in the graphene lattice it forms the intense binding energy between sodium ions in the electrolyte; these ions accommodate in the graphene lattice; it boosts the electrochemical utilization and enhances the pseudocapacitive property. The pseudocapacitive reactions [Eq. (5),(6)] occur through pyridinic N, pyrrolic N are located at the periphery of few-layered graphene lattice [49].



Presence of sulfur in the graphene lattice forms oxidized sulfur groups like sulfonic and sulfonate groups, which contribute to the pseudocapacitance. The presence of electron-rich Sulphur in aromatic lattice improves the polarization and subsequently increases the pseudocapacitive behavior. Electron-rich sulfur in aromatic lattice increases the total polarization of the aqueous medium under an applied electric field, subsequently increasing the dielectric con-

stant of the electrolyte, which enhances the capacitance and charge transfer kinetics. The pseudocapacitive behavior rises due to the series of redox faradic reactions provided by the additional sulfone and sulfoxide species attached to the aromatics of graphene [51,52]. Co-doped hetero atoms (S and N) in the graphene lattice forms hydrogen bond-like N-S-H bond, which also contributes to the electrochemical performance. Redox reactions occur as shown in Eqs. (7),(8) through the electron-rich sulfur present in the form of sulfone and sulfoxide in few-layered graphene lattices [49,50]. Eqs. (7), (8) show O=S=O and S=O bonds, respectively.



5. Conclusions

S, N-FLG is synthesized by using easy and scalable microwave irradiation of acid-treated graphite flakes. S, N-FLG electrodes show an energy density of 15 Wh kg⁻¹ at a power density of 298 W kg⁻¹ in 6M NaOH solution. The S, N-FLG electrodes show the specific capacitance of 298 F g⁻¹ at a current density of 1 A g⁻¹ and stable over 10,000 continuous charge-discharge cycles with 95% capacitance retention at 1 A g⁻¹. The charge storage process consists of both diffusion-controlled processes as well as surface-confined

charge transfer processes in S, N-FLG. Almost identical diffusion coefficients of D_{Na^+} , D_{OH^-} are the 3.31×10^{-11} , $3.29 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ respectively are obtained indicates cations, as well as anions, are equally penetrating to the electrode. The obtained capacitance is due to the maximum utilization of few-layered graphene sheets (less agglomeration), more wettability, highest intrinsic surface capacitance due to the synergetic effect of the formation of N-S-H hydrogen bonds, and S, N co-doping in graphene aromatic rings.

Declaration of Competing Interest

The authors declare no competing financial interest.

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Author contributions

The manuscript was written through the contributions of all authors. All authors have given endorsement to the final version of the manuscript.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.electacta.2020.137209](https://doi.org/10.1016/j.electacta.2020.137209).

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